

FORTIGATE SD-WAN DUAL ISP FAILOVER (METHOD OF PROCEDURE)

1. Purpose

The purpose of this MOP is to implement FortiGate SD-WAN within the existing FortiGate HA cluster to provide intelligent WAN path selection, improved Internet resiliency, and automatic failover between the Dedicated Internet Access (DIA) and Broadband (BB) Internet circuits.

This implementation enables:

- Active utilization of dual Internet providers
- Performance SLA-based WAN health monitoring
- Automatic failover during ISP outages
- Automatic restoration to the preferred DIA circuit following recovery
- Reduced operational impact during WAN failures
- Improved Internet availability for users within the HQ Network (10.42.0.0/16)
- Centralized WAN management through the FortiGate SD-WAN framework

This change will leverage the existing DIA circuit (172.16.0.1) and Broadband circuit (172.16.0.9) while introducing SD-WAN monitoring, traffic steering, and automated failover capabilities without requiring modifications to the underlying campus network architecture.

2. Scope

This procedure applies to the implementation of SD-WAN functionality on the existing FortiGate HA cluster and the integration of the existing DIA and Broadband Internet circuits into a redundant WAN architecture.

This procedure applies to:

FortiGate Infrastructure

- FG-FW-A (Primary)
- FG-FW-B (Secondary)

WAN Infrastructure

ISP-DIA

- WAN Network: 172.16.0.0/29
- ISP Gateway: 172.16.0.1
- FortiGate Interface: Port 8 (172.16.0.2)

ISP-BB

- WAN Network: 172.16.0.8/29
- ISP Gateway: 172.16.0.9
- FortiGate Interface: Port 9 (172.16.0.10)

SD-WAN Components

- WAN Member Configuration
- Performance SLA Monitoring
- Traffic Steering Policies
- Automatic WAN Failover
- Automatic WAN Recovery

HQ Network

- 10.42.0.0/16

Validation Targets

- 8.8.8.8
- 1.1.1.1

3. Pre-Implementation Requirements

Before beginning:

- Confirm administrative access to the FortiGate HA Cluster.
- Confirm FG-FW-A and FG-FW-B are healthy and synchronized.
- Confirm connectivity to ISP-DIA-DEMARC (172.16.0.1).
- Confirm connectivity to ISP-BB-DEMARC (172.16.0.9).
- Verify both ISP circuits provide Internet connectivity.
- Verify Internet access from the HQ-Network (10.42.0.0/16).
- Backup the current FortiGate configuration.
- Document existing WAN interfaces, static routes, and firewall policies.

4. Implementation Procedure

Step 1 – Enable SD-WAN

- Navigate to:
 - Network -> SD-WAN
 - Enable SD-WAN Functionality
 - Apply Changes
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Step 2 – Add WAN Members

- Navigate to:
 - Network -> SD-WAN -> SD-WAN Zones
 - Create New -> SD-WAN Zone
 - Name: “virtual-wan-link” (also created by default for members)
 - Return to SD-WAN Zones
 - Create New -> SD-WAN Member
 - Name: “to_DIA”
 - Interface: Port8
 - Gateway: 172.16.0.1
 - Priority: 1
 - Return to SD-WAN Zones
 - Create New -> SD-WAN Member
 - Name: “to_BB”
 - Interface: Port9
 - Gateway: 172.16.0.9
 - Priority: 2
 - Apply Changes
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Step 3 – Configure Performance SLA Monitoring

- Navigate to:
 - Network -> SD-WAN -> Performance SLAs
 - Create New
 - Name: “DIA_SLA”
 - Target Server: 8.8.8.8
 - Interface: to_DIA
 - Return to Performance SLAs
 - Create New
 - Name: “BB_SLA”
 - Target Server: 1.1.1.1
 - Interface: to_BB
 - Apply Changes
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Step 4 – Create SD-WAN Address Objects

- Navigate to:
 - Policy & Objects -> Addresses
 - Create New -> Address
 - Name: “DIA_TEST”
 - Type: Subnet
 - IP/Netmask: 8.8.8.8/32
 - Return to Addresses
 - Create New -> Address
 - Name: “BB_TEST”
 - Type: Subnet
 - IP/Netmask: 1.1.1.1/32
 - Return to Addresses
 - Create New -> Address
 - Name: “HQ-Network”
 - Type: Subnet
 - IP/Netmask: 10.42.0.0/16
 - Apply changes
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Step 5 – Create SD-WAN Traffic Steering Rules

- Navigate to:
 - Network -> SD-WAN -> SD-WAN Rules
 - Create New
 - Name: “DIA-Steering”
 - Destination: DIA_TEST
 - Outgoing Interfaces: Manual
 - Interface Preference: “to_DIA” first, then “to_BB”
 - Return to SD-WAN Rules
 - Create New
 - Name: “BB-Steering”
 - Destination: BB_TEST
 - Outgoing Interfaces: Manual
 - Interface Preference: “to_BB” first, then “to_DIA”
 - Apply Changes
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Step 6 – Configure Internet Access Policy

- Navigate to:
 - Policy & Objects -> Firewall Policy
 - Create New
 - Name: “HQ-Network -> SD-WAN”
 - Incoming Interface: port2
 - Outgoing Interface: virtual-wan-link
 - Source: HQ-Network
 - Destination: all

- Service: ALL
 - NAT: Enabled
 - IP Pool Configuration: Use Outgoing Interface Address
 - Log Allowed Traffic: All Sessions
 - Capture Packets: ✓
 - Apply Changes
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5. Validation Plan

Step 1 – Verify SD-WAN Status

- Navigate to:
 - Network -> SD-WAN
 - Confirm:
 - SD-WAN = Enabled
 - “virtual-wan-link” exists
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Step 2 – Verify WAN Members

- Navigate to:
 - Network -> SD-WAN Zones
 - Confirm:
 - to_DIA = up
 - to_BB = up
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Step 3 – Verify Performance SLA Status

- Navigate to:
 - Network -> SD-WAN -> Performance SLA
 - Confirm:
 - DIA_SLA = Healthy
 - BB_SLA = Healthy
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Step 4 – Verify DIA Traffic Steering

- From any host/appliance in HQ-Network, generate traffic to 8.8.8.8 (ping)
 - Navigate to:
 - Network -> SD-WAN -> SD-WAN Rules
 - Confirm: DIA-Steering rule match counter increases
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Step 5 – Verify BB Traffic Steering

- From any host/appliance in HQ-Network, generate traffic to 1.1.1.1 (ping)
 - Navigate to:
 - Network -> SD-WAN -> SD-WAN Rules
 - Confirm: BB-Steering rule match counter increases
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Step 6 – Verify WAN Path Selection

- Navigate to:
 - Log & Report -> Forward Traffic
 - Confirm for traffic destined to 8.8.8.8:
 - SD-WAN Rule: DIA-Steering
 - Outgoing Interface: to_DIA
 - Confirm for traffic destined to 1.1.1.1:
 - SD-WAN Rule: BB-Steering
 - Outgoing Interface: to_BB
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Step 7 – Verify Automatic Failover

- Disconnect DIA connectivity
 - Navigate to:
 - Network -> SD-WAN -> Performance SLAs
 - Confirm DIA_SLA = Dead/Down
 - Generate traffic to: 8.8.8.8
 - Navigate to:
 - Network -> SD-WAN -> SD-WAN Rules
 - On DIA-Steering rule:
 - Confirm Member primary switches from “to_DIA” to “to_BB”
 - Confirm rule match counter increase
 - Confirm traffic to 8.8.8.8 remains reachable after failover
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Step 8 – Verify Automatic Recovery

- Restore DIA connectivity
- Navigate to:
 - Network -> SD-WAN -> Performance SLAs
 - Confirm DIA_SLA = Healthy
- Generate traffic to: 8.8.8.8
- Navigate to:
 - Network -> SD-WAN -> SD-WAN Rules
 - On DIA-Steering rule:
 - Confirm Member primary switches from “to_BB” back to “to_DIA”

- Confirm rule match counter increase
 - Confirm traffic to 8.8.8.8 remains reachable after recovery
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Step 9 – Reverse Failover Testing

- Repeat steps 7 and 8 with the following changes:
 - Disconnect BB connectivity
 - Check BB_SLA
 - Generate traffic to 1.1.1.1
 - Verify BB-Steering rule
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6. Rollback Procedure

Rollback should be performed if:

- Internet connectivity is lost.
 - Performance SLA monitoring fails.
 - Incorrect WAN path selection occurs.
 - SD-WAN traffic steering does not function correctly.
 - Automatic failover does not occur as expected.
 - Automatic recovery does not occur as expected.
 - Validation testing cannot be successfully completed.
 - Unexpected routing behavior is observed
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Step 1 – Remove SD-WAN Traffic Steering Rules

- Navigate to:
 - Network -> SD-WAN -> SD-WAN Rules
 - Delete:
 - DIA-Steering
 - BB-Steering
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Step 2 – Remove Address Objects

- Navigate to:
 - Policy & Objects -> Addresses
 - Delete:
 - DIA_TEST
 - BB_TEST
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Step 3 – Remove Performance SLA Monitoring

- Navigate to:
 - Network -> SD-WAN -> Performance SLAs
 - Delete:
 - DIA_SLA
 - BB_SLA
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Step 4 – Remove SD-WAN Members

- Navigate to:
 - Network -> SD-WAN -> SD-WAN Zones
 - Remove:
 - to_DIA
 - to_BB
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Step 5 – Remove SD-WAN Internet Policy

- Navigate to:
 - Policy & Objects -> Firewall Policy
 - Delete:
 - HQ-Network -> SD-WAN
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Step 6 – Disable SD-WAN

- Navigate to:
 - Network -> SD-WAN
 - Disable:
 - SD-WAN
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Step 7 – Restore Previous Routing Configuration

- Navigate to:
 - Network -> Static Routes
 - Restore the pre-change default routing configuration.
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Step 8 – Validate Rollback

- Verify Internet connectivity from the HQ-Network
- Verify connectivity to:
 - 8.8.8.8
 - 1.1.1.1

Expected result:

- SD-WAN functionality has been removed.
 - Internet connectivity has been restored to the pre-change state
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